Introduction to Economics

Class 11 — Production and costs

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Plan for today

Warm-up
Electricity for nothing?
What counts as a cost; in pairs: a yoga studio
One press, more printers
Total, average and marginal cost
In pairs: an ice-cream maker's costs
Break
Electricity for nothing: why
The U-shaped cost curves; in pairs
Short run and long run: one press or two
Economies of scale, and why they end
In pairs: a laundry grows
In one line; before Monday 2 November

Slide 1 Title

Preview version. The full slides and handout are posted after class.

Block 4's first class. Classes 8 to 10 looked at whole markets: surplus and taxes, controls and trade, and the cases where a market gets it wrong. From today the camera moves inside one firm: what it costs to make things, how that cost changes as the firm makes more, and what changes once it can grow. One small firm, a print shop, carries the whole class; a real price that fell to zero opens it, and the day's tools answer it after the break.

Slide 2 Plan for today

The class at a glance. Before the break, one small firm from its bills to its cost per bag: what economists count as a cost, how output grows as workers are added to a single press, and the three measures of cost that follow from that. After the break, the pictures those numbers make, and what changes when the firm can change its size as well as its staff.

Two halves of fifty minutes with a ten-minute break; the opening question about electricity is answered in the second half.

Warm-up: what does the concert cost you?

Tonight: a concert or your supermarket shift?

Ticket paid: 30, no refund · shift pays 45 · bus to the concert: 2

1. Your opportunity cost of going?
2. A friend offers 40 for the ticket. Now?
3. The concert is worth 60 to you. Go?

In pairs, two minutes; then hands per item.

Slide 3 Warm-up: what does the concert cost you?

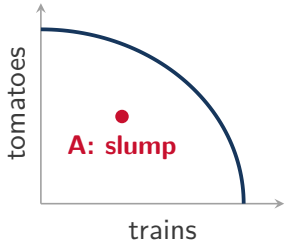
A warm-up on Class 2's opportunity cost, with a concert against a supermarket shift. Before class, decide which of the numbers on the slide you actually give up by going, and whether a ticket you already own can still cost you something once a friend wants it; then compare what the evening is worth to you with what it costs.

Revealed in class.

Warm-up: what moves the frontier?

A country makes trains and tomatoes

1. A greenhouse method doubles the tomatoes per worker
2. Workers arrive who can do either job
3. Factories idle in a slump reopen



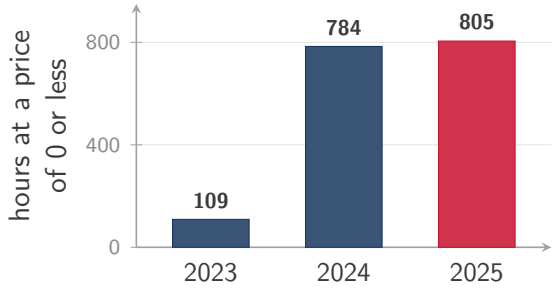
In pairs, one minute; then hands.

Slide 4 Warm-up: what moves the frontier?

The second warm-up is Class 3's frontier, on trains and tomatoes. Before class, ask for each event whether it changes the most the country could make of trains, of tomatoes, of both or of neither; for the slump, look at where point A sits on the small figure before deciding what reopening the factories does.

Revealed in class.

Spain, 2025: about 800 hours of electricity at a price of zero or less



Almost all in the middle of the day, most in spring.

Why would anyone sell electricity for nothing?

Own count from the day-ahead prices of OMIE (the Spanish–Portuguese wholesale electricity market), mainland Spain (set per quarter-hour since October 2025: four of them count as one hour).

Slide 5 Spain, 2025: about 800 hours of electricity at a price of zero or less

The hook, counted from the Iberian market operator's own price files. OMIE runs the market where Spain's and Portugal's generators sell the next day's electricity, hour by hour (quarter-hour by quarter-hour since October 2025), and in 2025 the price for mainland Spain was zero or below in about 805 of the year's hours, roughly one hour in eleven, almost all of them around midday and most of them in spring; in 2023 it had happened in 109. Your bill at home was not zero, because it adds the grid, taxes and charges; what fell to nothing was the wholesale price. The question is the slide's: why would a firm keep selling its product for nothing, or even pay to sell it? Keep it open; slide 14 answers it with the day's tools.

Counting what the owner gives up: the shop earns 1,700, not 4,400

The print shop's month

20 days · 4 printers · 220 bags a day at 4 euros

	euros
sales	17,600
− explicit costs (rent and press, printers)	13,200
= accounting profit	**4,400**
− implicit costs (her pay at an agency, interest forgone)	2,700
= economic profit	**1,700**

Accountants count the money paid out; economists count every cost given up.

Slide 6 Counting what the owner gives up: the shop earns 1,700, not 4,400

Meet the firm the whole class follows: a print shop with one press, printing customers' designs on bags they bring. The table counts its month twice. The accountant subtracts only money paid out; the economist also subtracts what the owner gives up by being here, the pay she could earn elsewhere and the interest her savings could earn invested elsewhere, which is why the reading says a firm's costs are its opportunity costs. The smaller profit is the one that says whether the shop is worth running: positive, so it beats her best alternative. The savings themselves are not a monthly cost; only the interest they could earn is.

In pairs, three minutes: a yoga studio's profit

A yoga studio's year

sales	72,000
rent, teachers, heating	40,000

The owner left a job paying 29,000. Her 75,000 of savings in the studio could earn 4 % a year.

1. Accounting profit?
2. Implicit costs?
3. Economic profit?
4. Economic profit 0: is she losing money?

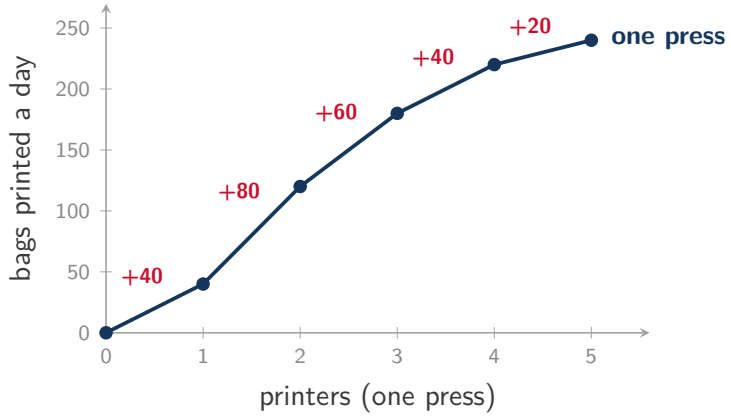
Slide 7 In pairs, three minutes: a yoga studio's profit

Exercise 1 is slide 6 on another firm. Before class, work out what the owner's accountant would report from the table alone, then list what she gives up that nobody invoices and ask whether the savings themselves belong on that list; finally ask what an economic profit of exactly zero would mean for her.

Revealed in class.

One press, more printers: from the third on, each adds fewer bags

A print shop in Poblenou: one press; customers bring the bags.

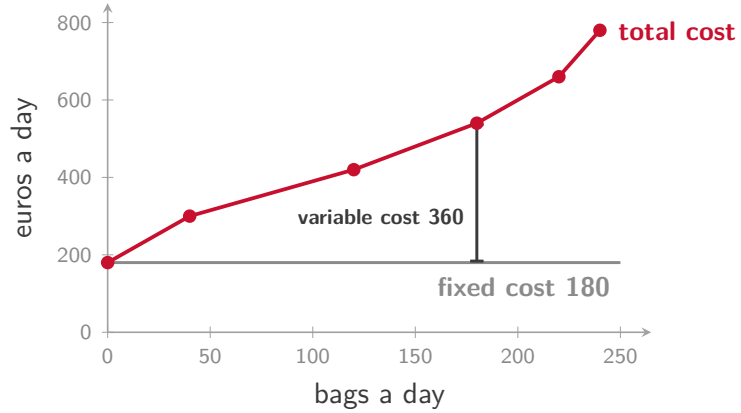


Marginal product: the bags one more printer adds. It falls once they crowd the press.

Slide 8 One press, more printers: from the third on, each adds fewer bags

The shop's production function: how many bags a day each number of printers can make with the press and the workshop fixed. What matters is not the total but the step between the points, the extra bags one more printer adds. It rises at first, because two people can split the work, and then falls, because every extra printer waits for the same press and gets in the others' way; that fall is what the reading calls diminishing marginal product, and it arrives in any firm that adds one input while another stays fixed. Total output never falls here; it is the added bags that shrink, which is the slip to avoid.

Total cost climbs faster as the press gets crowded



From the third printer on, each adds 120 but fewer bags: the curve gets steeper.

Slide 9 Total cost climbs faster as the press gets crowded

The same production function turned into money. Rent and the press cost the same whether the shop prints nothing or 240 bags, so their line is flat; each printer adds the same amount to cost, so total cost climbs by equal steps as printers are added, but from the third printer on each step buys fewer bags than the last, which is why the curve bends upward to the right. The gap between the two lines at any output is the variable cost, the printers' pay. A flattening production function and a steepening cost curve are one fact seen from two sides. Fixed does not mean fixed for ever: it means not changing with output; the landlord can still raise the rent next year.

Per bag: the average bottoms out at 3.00; once the press is crowded, extra bags get dearer

printers	bags a day	total cost TC	AFC FC ÷ bags	AVC VC ÷ bags	ATC TC ÷ bags	MC extra cost ÷ extra bags
0	0	180	—	—	—	
1	40	300	4.50	3.00	7.50	3.00
2	120	420	1.50	2.00	3.50	1.50
3	180	540	1.00	2.00	**3.00**	2.00
4	220	660	0.82	2.18	**3.00**	3.00
5	240	780	0.75	2.50	3.25	6.00

FC, VC: fixed, variable cost · AFC, AVC, ATC: average fixed, variable, total cost · MC: marginal cost

Fixed cost 180 a day · each printer 120 a day.

MC = 120 ÷ the extra bags: it rises as each printer adds fewer.

Slide 10 Per bag: the average bottoms out at 3.00; once the press is crowded, extra bags get dearer

The two questions a firm asks about cost, side by side: what the typical bag costs, and what one more costs. The average has two parts pulling opposite ways, the rent spread over ever more bags and the printers' pay per bag, which falls while teamwork helps and rises once the press is crowded; the sum falls, then climbs, and its lowest value marks what the reading calls the efficient scale. The last column is not total cost divided by bags but the rise in total cost divided by the extra bags it buys; since every printer costs the same, that is his pay divided by his marginal product, which is the link back to slide 8. Where the extra bags cost exactly the average, the average stands still, which is why its lowest value appears twice.

In pairs, four minutes: an ice-cream maker's costs

An ice-cream maker's day

tubs a day	total cost
0	60
10	100
20	130
30	180
40	260

1. Its fixed cost?
2. ATC at 30 tubs?
3. MC of the tubs from 30 to 40?
4. The rent rises by 30 a day. ATC at 30? MC from 30 to 40?

Slide 11 In pairs, four minutes: an ice-cream maker's costs

Exercise 2 is slides 9 and 10 on a table with five rows. Before class, read the fixed cost off the table, then compute the average at one row and the marginal cost between two rows, remembering what the marginal cost is divided by; for the last item, redo the table with the higher rent and compute both numbers again.

Revealed in class.

Three honey farms: how much does the market supply? (optional)

Jars a week each farm would sell

	at 6 euros	at 8 euros
Farm A	300	400
Farm B	0	250
Farm C	0	150

1. Market supply at 8 euros?
2. At 6 euros?
3. A fourth farm sells 100 at either price. At 6?

Slide 12 Three honey farms: how much does the market supply? (optional)

An optional exercise on Class 5's market supply, tied to what firms decide to sell. Before class, decide what has to be added to get a market quantity at one price and what must not be, and what a farm that would sell nothing at a price adds to the total there.

Revealed in class.

Break

Ten minutes

slide 13 of 26

Slide 13 Break

Ten minutes. After the break: the answer to the electricity question, then the print shop's costs as curves.

Electricity for nothing: why?

Revealed in class

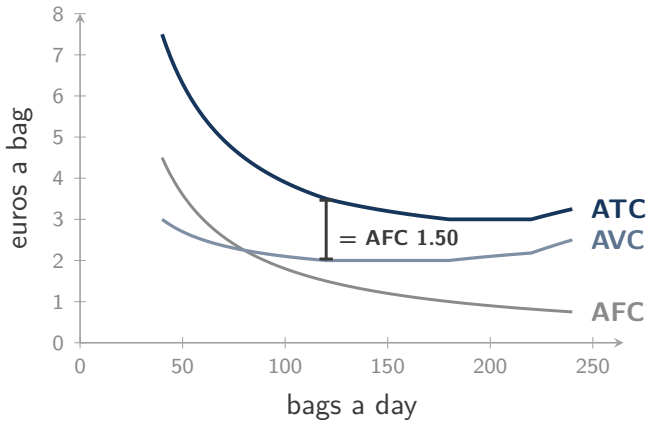
Think back to slides 9 and 10: which of a solar farm's costs change if it switches off for an afternoon?

slide 14 of 26

Slide 14 Electricity for nothing: why?

Revealed in class: which of a solar farm's costs change if it switches off for an afternoon? Think back to slides 9 and 10 before class.

Fixed cost spread thinner, a crowded press: average total cost is a U

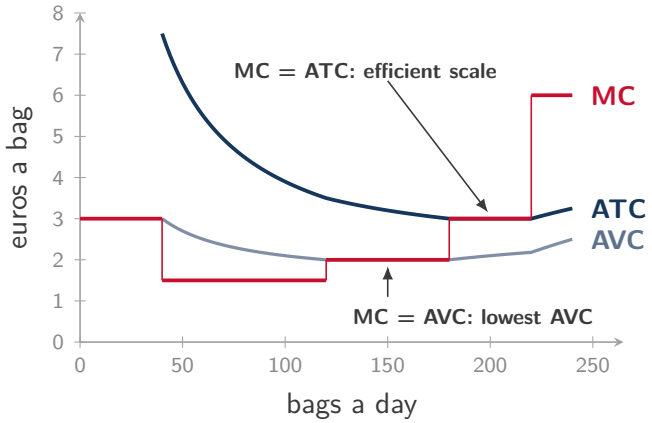


ATC = AFC + AVC: falling fixed cost per bag, then a crowded press.

Slide 15 Fixed cost spread thinner, a crowded press: average total cost is a U

Slide 10's table as curves, one at a time. The grey curve only ever falls, because the same rent is divided by more and more bags; the lighter blue one falls while the second printer's teamwork helps, runs flat, then rises as the press gets crowded. Average total cost is the two stacked, which is what the bracket shows at one output, so it is high at first because the rent sits on few bags and high again at the end because every bag needs more printer time; the U is the two forces taking turns. Our shop's bottom is flat because printers come in whole days; with half-days it would be rounder.

Marginal cost cuts through both averages at their lowest point



MC below the average pulls it down; above, pushes it up.

Slide 16 Marginal cost cuts through both averages at their lowest point

The last curve, and the rule that fixes where it crosses the others. Marginal cost falls while teamwork helps and then climbs steeply as the press gets crowded, the mirror of slide 8. Wherever the extra bag costs less than the current average, adding it drags the average down, and wherever it costs more, adding it lifts the average; so the average can only stop falling where the extra bag costs exactly the average, and that is where the two curves meet, at the bottom of each U. Think of a grade average and a new mark. The crossing with average total cost marks the efficient scale from slide 10; the same logic puts the other crossing at the bottom of average variable cost.

In pairs, two minutes: averages and the next one

Below the average or above it?

1. Average grade 6.0 after four exams; the fifth: 8.0. New average?
2. A firm's ATC is 6.00 at 100 units; the next 20 cost 3.00 each. ATC at 120?
3. Where MC equals ATC, ATC is...

Then hands per item.

Slide 17 In pairs, two minutes: averages and the next one

Three quick checks of slide 16's rule, two with numbers and one in words. Before class, work each average from a total rather than from the old average and the new number, and for the firm be careful what the new total is divided by; for the third item, say in words what slide 16's crossing point means.

Revealed in class.

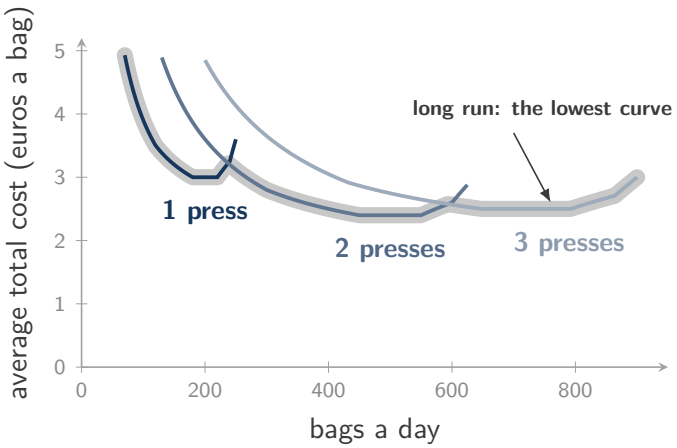
In the long run every cost can change: the shop can rent a second press

- Short run: the press is fixed; only printers change
- Long run: a second press, a bigger workshop
- In the long run every cost is variable
- How long? A print shop: months; a car plant: years

Slide 18 In the long run every cost can change: the shop can rent a second press

A definition slide, in words. The short run is however long some input stays fixed, here the press and the lease; the long run is long enough to change every input, so a cost that is fixed this year, the rent, becomes a choice in the long run. The split is not a number of months: it depends on how long the fixed input takes to change, which is why it is short for a print shop and years for a car plant. Everything before this slide was short run; the next three slides let the shop choose its size.

The long-run cost curve: the cheapest size for each output



In the long run the shop picks its size: the lowest curve at each output.

Slide 19 The long-run cost curve: the cheapest size for each output

Slide 15's U again, on a wider scale, next to the U the shop would have with two presses and with three. Each curve is a short run, a size the shop is stuck with; the long run is the choice among them, so for each output the shop would pick the size whose curve is lowest there, and the grey band traces those lowest pieces. Reading it: an order one press cannot fill at all becomes affordable with two, and a bigger size is only worth it once output is large enough to use it. With many sizes to choose from the scallops smooth into one flatter U, the reading's picture. The band falls, runs flat and rises again: the long run does not make cost fall for ever.

Double every input: if output more than doubles, the cost per bag falls

Now: 1 press, 4 printers: 220 bags for 660 a day, 3.00 a bag

Double every input: 2 presses, 8 printers, 1,320 a day.

if it makes	cost per bag	
550 bags	2.40	**economies of scale**
440 bags	3.00	**constant returns to scale**
330 bags	4.00	**diseconomies of scale**

What counts: does output grow faster than cost?

Slide 20 Double every input: if output more than doubles, the cost per bag falls

The long run in numbers, with the three names the reading gives. The shop doubles every input, so with the same input prices its cost doubles too; what decides the name is the race between output and cost, whether output grows faster, exactly as fast or slower, and the three rows show one case each. The cost per bag falls, stays or rises accordingly. The slip to avoid is reading economies of scale as total cost falling: total cost has doubled in every row; it is the cost per bag that moves. Slide 19's falling band was the first row; its rising end was the third.

Bigger is cheaper per bag — until coordination gets hard

Bigger, cheaper

workers who each do one task
bulk buying: ink by the barrel
machines that only come big
cheaper loans

Too big, dearer

more managers, slower decisions
messages lost on the way
workers far from the big picture

Slide 21 Bigger is cheaper per bag — until coordination gets hard

The reasons behind slide 20's names. Size lowers the cost per unit when it lets workers specialise, buys inputs in bulk, uses machines that only exist in big sizes and borrows more cheaply; the reading adds the economies a whole industry gets from clustering in one place. Size raises it when the firm becomes hard to run: more layers, slower decisions, messages lost between departments, workers who never see the finished product. Both cards hold for the same firm at different sizes, which is why the long-run curve of slide 19 first falls and later rises, and why a bigger firm is not always a cheaper one.

In pairs, five minutes: a laundry grows

A laundry for hotels New machines take months to arrive.

	machines	staff	kilos a day	cost a day
now	4	2	300	600
doubled	8	4	750	1,200
doubled again	16	8	1,200	2,400

1 ATC now?

2 Doubled: ATC? What kind of returns?

3 Doubled again: ATC? Returns?

4 A hotel adds 100 kilos a day from next week. What can change by then?

Slide 22 In pairs, five minutes: a laundry grows

Exercise 3 is slides 18 to 20 on a laundry that doubles twice. Before class, compute the cost per kilo at each of the three sizes, then compare how kilos and cost each grew between one size and the next and name the kind of returns from that comparison rather than from the level the average ends at; for the last item, ask what the laundry can and cannot change within a week.

Revealed in class.

Tote bags: which curve moves? (optional)

The market for printed tote bags

1. Printers' pay rises
2. The charge for a plastic bag doubles
3. Demand rises and totes get dearer: does supply shift too?

Slide 23 Tote bags: which curve moves? (optional)

An optional exercise on Class 5's shift question, tied to slide 9's input costs. Before class, ask for each event whether it changes what the printers would offer at every price, what buyers would want at every price, or only the price itself, and then which curve, if any, moves and which way.

Revealed in class.

Cheap solar: positive or normative? (optional)

Cheap solar: three statements

1. "More solar panels bring more hours at a price of zero."
2. "Electricity should be cheap for everyone."
3. A and B want cheap electricity. B: "Slow down: zero prices will scare off investors."
   B's reason: values or a prediction?

Slide 24 Cheap solar: positive or normative? (optional)

An optional exercise on Class 4's distinction, on the price from slide 5. Before class, ask of each statement whether evidence could in principle show it to be wrong; for the third, ask whether B disagrees with the goal or with a forecast.

Revealed in class.

A cost is what the firm gives up, paid out or not.

Once workers crowd the press, each adds less, so marginal cost rises; it cuts average cost at its lowest point.

In the long run every cost can change: economies of scale, then diseconomies.
Monday: how much a firm sells at a given price.

Before Monday 2 November

- Monday: perfect competition; monopoly begins
- Read Mankiw & Taylor: Ch. 5, pp. 123–137
- Problem set 2 (Block 3): due Monday 2 November, on paper, in class
- Problem set 3 (Block 4): handed out Monday; due Thursday 12 November
- Practise in Sextant: the units the course page lists

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Slide 25 In one line

The class in four lines, to read slowly: the first is slides 6 and 7, the second slides 8 to 10 and 16, the third slides 18 to 21, and the fourth names Monday.

Slide 26 Before Monday 2 November

Monday 2 November: perfect competition, how much a firm like the print shop sells when it cannot set the price, and the start of monopoly. Reading: Mankiw & Taylor chapter 5, pages 123 to 137. Problem set 2, on Block 3, is due on Monday, on paper, in class; Problem set 3, on Block 4, is handed out on Monday and due on Thursday 12 November. Practise in Sextant with the units the course page lists.

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